

Kimberly Palaguachi-Lopez

kp2809@cumc.columbia.edu | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Columbia University Mailman School of Public Health – New York, NY

May 2026

Master of Science in Biostatistics, Health Data Science Track

- Relevant Coursework: Biostatistics I & II, Epidemiology, Probability, Statistical Inference, SAS, Data Science I & II, Survival Analysis, Longitudinal Data Analysis, Randomized Clinical Trial Methodology, SQL for Relational Databases.

Lehigh University – Bethlehem, PA

May 2024

Bachelor of Science in Population Health

- Relevant Coursework: Population Health Research Methods, Methods in Epidemiology, Population Health Data Science & Algorithms Labs I & II, Spatial Epidemiology, Environmental Epidemiology, Outbreak Science & Disease Forecasting, Application of Statistical Inference.

RESEARCH EXPERIENCE

Boehringer Ingelheim Pharmaceuticals – Remote

August 2025 – Present

Clinical Data Science – Statistics Co-op MS

- Supported clinical validation and statistical analyses across phase I-III studies by processing and analyzing clinical databases using SDTM and ADaM datasets to generate regulatory-quality tables, listings, and figures from SAS to R.
- Data manipulation of table functions in R to generate formatted tables displaying descriptive statistics and endpoint analysis of data to support statistical analysis plans (SAPs); recorded in Analysis Data Reviewer's guide.
- Developed interactive user-friendly R Shiny dashboards to modernize reporting of statistical analysis and visualize key data outcomes for cross-functional teams and CRO partners, improving efficiency in regulatory reporting.

Joslin Diabetes Research Center – Boston, MA

June 2025 – August 2025

Summer Research Intern

- Conducted longitudinal analysis of NIH phase II clinical trial data following an ITT protocol using SAS and linear mixed-effect models to evaluate longitudinal kidney function endpoints in Type 1 Diabetes patients.
- Performed longitudinal biomarker analyses and data validation by comparing multiple renal biomarkers (creatinine, cystatin-C, and direct GFR) to ensure data integrity and accuracy in clinical endpoints.
- Synthesized and presented findings on treatment effect in kidney function, generating evidence to inform future large-scale research studies and population health interventions aimed at delaying kidney decline in T1D.

Breakthrough T1D Research Foundation | Formerly JDF – New York, NY

June 2024 – August 2024

Improving Lives Research Internship

- Analyzed 200+ clinical trials in T1D and T2D cardiovascular disease research, identifying key gaps in biomarker utilization by conducting a comprehensive clinical trial landscape analysis.
- Researched biomarker development for cardiovascular disease and heart failure to support early detection strategies for young T1D patients by assessing recent literature and emerging diagnostic tools.
- Presented a 30-minute landscape analysis to the research department by synthesizing findings and proposing 3 promising cardiovascular drug treatments for future grant application strategies.

College of Health at Lehigh University – Bethlehem, PA

January 2024 – May 2024

Epidemiology Statistical Analysis Assistant

- Processed and merged 300+ participant baseline and follow-up data, and performed exploratory analysis.
- Conducted descriptive statistics and performed hypothesis testing, identifying significant changes in health scales over time, by analyzing cross-sectional survey responses.
- Co-authored a manuscript selected for presentation at APHA 2024 conference contributing to public health discourse, by synthesizing findings and collaborating with faculty on statistical interpretation.

STEM – Summer Institute at Lehigh University – Bethlehem, PA

June 2023 – September 2023

Summer Research Student

- Processed 12 years (2011-2023) of Mortality Morbidity Weekly Report (MMWR) Influenza A/B data across 67 Pennsylvania counties identifying seasonal flu trends, by developing heat maps for data visualization using Python.

- Built an automated data pipeline to streamline CDC data to epidemiological modeling workflow and estimated parameter values for an Agent-Based Model (ABM) simulated in Netlogo.
- Evaluated the effectiveness of the ABM for predictive accuracy by analyzing disease forecasting and refined simulation parameters.

College of Health at Lehigh University– Bethlehem, PA

September 2022 – December 2022

Co-Research Author

- Conducted extensive literature review research on the relationship between asthma and cold temperatures.
- Contributed to the drafting and publication of an academic research paper focused on nasal malondialdehyde mediating the effects of cold air exposure on pediatric asthma symptoms.

Institute of Critical Race & Ethnic Studies at Lehigh University – Bethlehem, PA

February 2021 – May 2021

Undergraduate Data Analyst Assistant

- Attended weekly 3 hours of meetings with colleagues to review agendas and planning new projects.
- Gather 5,000+ participant's personal experiences into Google Sheets during the COVID-19 pandemic via social media, used to determine how the pandemic is affecting race and gender in data analysis.

WORK EXPERIENCE

Shift's Chat Bot Advising Co-Op, – Remote

November 2022 – December 2023

KIPP NYC Representative

- Engaged in strategic meetings with partner organizations to offer expert consultation on testing procedures, while also delivering valuable feedback to enhance their testing initiatives and improve high school student alumni support through college transition.
- Facilitated effective communication and collaboration between students, caregivers, and the KIPP NYC community, fostering a cohesive partnership within the hybrid advising co-op network.
- Actively participated in monthly advisory council meetings and co-op webinar calls to assess the outcomes of chatbot testing, ensuring that data-driven insights were leveraged for continuous improvement.

Village Care Max – Manhattan, NY

June 2023 – August 2023

Data Entry Intern

- Retrieved vaccination Electronic Health Record (EHR) data from the Citywide Immunization Registry for Village Care patients across the broader New York City area.
- Systematically processed and incorporated new data into database systems, cross-referenced and validated against Village Care records.
- Proficiently managed, organized, and securely stored over 3,000 data entries within spreadsheets, consistently upholding strict HIPAA compliance standards.

Pathways To Trust – Remote

July 2022 – August 2022

Strategic Partnership Donor Development Intern

- Conducted comprehensive research to identify prospective donors and sponsors for Pathways To Trust, facilitating the organization's patient advocacy programs
- Collaborated with PTT partners to actively seek out suitable research centers for disease clinical trials, contributing to the expansion of the organization's participant network.
- Orchestrated effective communication and teamwork with a group of interns to proactively explore grant opportunities for PPT to ensure eligibility for future funding.

PUBLICATIONS

Health Effects of a Natural Experiment Introducing Nature Window Views Among Healthcare Workers

Submitted to Health Environments Research & Design Journal

November 2025

16 Pages, Co- authors: Fry, Dustin; Ganns, Elyse; Palaguachi-Lopez, Kimberly; Eaves-Lyter, Emily; McIntire, Russell

Nasal Malondialdehyde Mediating the Effects of Colder Air Exposure on Pediatric Asthma Symptoms

Pediatric Research

April 2024

23 Pages, Co- authors: He, Linchen; Norris, Christina; Palaguachi-Lopez, Kimberly; Barkjohn, Karoline; Li, Zhen; Li, Feng; Zheng, Yinping; Black, Marilyn; Bergin, Michael; Zhang, Junfeng (Jim)

CONFERENCE ABSTRACTS

Are Greener Window views associated with better mental health among workers at a rehabilitation hospital? A natural experiment

APHA 2024 Annual Meeting and Expo

November 2024

Co-authors: McIntire, Russell; Palaguachi-Lopez, Kimberly; Rafai, Nader; Lyter, Emily; Kolakowsky-Hayner, Stephanie

LEADERSHIP ACTIVITIES

R.I.S.E Mentorship Columbia University – New York, NY

August 2025 – Present

Mentor

- Mentoring three first-year M.S. Biostatistics students to support their transition into graduate-level coursework, statistical programming, and research opportunities.
- Facilitating connections to faculty-led research projects, departmental seminars, and professional development workshops to enhance mentees' academic growth and research engagement.
- Collaborating with program administrators to monitor mentees' progress and tailor mentorship strategies to academic and professional goals.

The Braven Accelerator – Remote

August 2023 – December 2023

Fellowship

- Developed workforce skills such as leadership, operating and managing, communicating & networking, working in teams, and problem solving by completing a 15-week leadership and professional development course.
- Facilitated data collection as lead researcher in the capstone group project where we aim to increase United Airlines flight travel for non-frequent air traveler college students.

Southeast Asia at Lehigh (SEAL) – Bethlehem, PA

August 2022 – May 2024

Vice President

- Promoted from club secretary to club vice president based on a demonstrated commitment to the organization and my leadership contributions during a critical period of need.
- Facilitated and planned many Southeast Asian cultural events while creating an inclusive environment for students.
- Provided dynamic leadership and strategic vision for the SEAL club executive board and sub-committee by improving event logistics for maximum event efficiency contributing to being awarded Student Organization of 2023-2024.

F1RST Student Organization – Bethlehem, PA

August 2022 – May 2024

- Mentored first and second year first generation students on academic planning, professional development, and navigating university resources to support their transition to college.

HONORS & AWARDS

Dean's List– Lehigh University

Spring 2021- Spring 2024

Recognized outstanding academic achievement who have earned a scholastic average of 3.6, or greater; and carry at least 12 hours of regularly graded courses.

Hispanic Scholarship Fund Scholar

June 2024

Selected as one of 10,000 scholars from a competitive pool of over 100,000 applicants. The distinction recognizes academic excellence, hard work, and dedication.

Tri Alpha Honor Society Member– Lehigh University

March 2024

The Alpha Alpha Alpha (Tri-Alpha) Epsilon Chapter at Lehigh recognizes the academic achievements of first-generation college students who have achieved an overall undergraduate GPA of at least 3.5 on a 4.0 scale.

SKILLS

Statistical Languages: Proficient in SAS, R, Python, and SQL for relational databases.

Data Management: Relational databases, data extraction and cleaning, healthcare data (relational/hierarchical).

Statistical Analysis: Survival analysis, power & sample size calculation, longitudinal data & multivariate analysis.

Data Analysis: Skilled in Python libraries (NumPy, Pandas, Matplotlib, SciPy), supervised/unsupervised machine learning techniques, model training, prediction modeling, and statistical modeling; GitHub for version control and collaboration.

Data visualization: SAS Graph, R Packages (ggplot2, plotly, R Shiny, Tplyr, phamaRTF), ArcGISPro mapping.